Do Now · retrieval

Synoptic Revision 2 — Final Test Preparation

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

A username must contain at least one character. Give an example of invalid input and explain what the program should do.

Q2

For a full adder with A = 1, B = 0 and CarryIn = 1, calculate Sum and CarryOut.

Q3

A loop starts with total = 0 and adds the values 2, 4 and 6 once each. What is total after the third iteration?

Q4

A half adder has inputs A and B. Its Sum output is A XOR B and Carry output is A AND B. For inputs (A,B) = (0,1), (1,0) and (1,1), calculate both outputs and explain why the final case produces a carry.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Do not copy corrections — CODE:
1. Attempt from memory
2. Check mark scheme
3. Write why the mark was lost
4. Retry without looking — The retry proves the correction has stuck.

Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

normalised — A floating-point form using the available mantissa precision efficiently.

pointer — A location/index/address linking to another item.

LIFO — Last In, First Out.

FIFO — First In, First Out.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Use the mark value

Follow the worked method

1. Direction — CODE:
right shift

2. Amount — CODE:
two places

3. Final answer — CODE:
Shift right by two places.

Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Complete both parts
CODE:
result = ?
effect = ?

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Data Types — OCR practice

Complete the Teams questions on negative binary, floating point and bit shifts. Mark them immediately.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 2

Personal checklist

Write the two exact things you will check before handing in the test next lesson.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Can you name the two exact things you will check before handing in the test?

Use key vocabulary from memory

1. Explain normalised in your own words.

2. Explain pointer in your own words.

3. Explain LIFO in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.